🏥 ArogyaLink — Technical Design Document (TDD)

Diagnostic Lab Automated Report-Delivery Portal for Patients and Referring Doctors

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Client: Company A, Coastal Andhra Pradesh Diagnostic Network (Vijayawada • 6 Central Labs & 38 Centres)
Public GitHub Repository: https://github.com/danimadhavapuri/ArogyaLink.git
Live Web Application: https://arogyalink-sc22.onrender.com

1. Executive Summary & Problem Context

Company A operates a diagnostic chain with 6 central labs and 38 outpatient collection centres across Vijayawada and coastal Andhra Pradesh. Their existing workflow involved 260 staff using 9 shared phones to manually dispatch unencrypted signed PDF reports over WhatsApp and SMS.

This process incurred significant operational friction and legal risks under the **Digital Personal Data Protection (DPDP) Act 2023**: reports remained permanently in chats, 15 reports a month were misdirected to wrong numbers, patients overwhelmed front-desks with status inquiries (40 staff hours/day), and critical panic results (e.g. Potassium 6.8 mmol/L) went unintercepted.

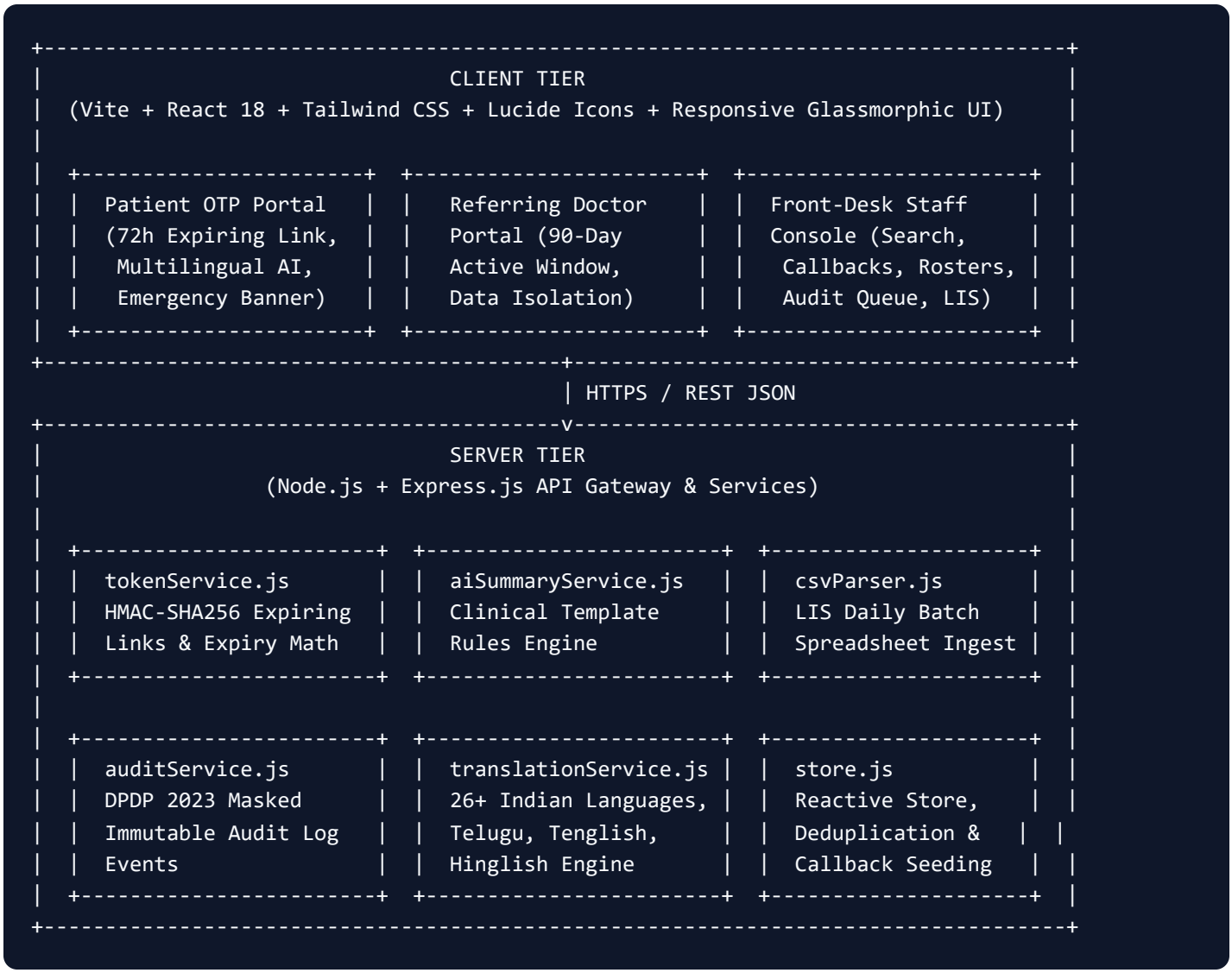
ArogyaLink introduces a zero-install, secure web portal governed by 72-hour signed expiring links, mobile OTP authentication, deterministic clinical AI summaries in 26+ Indian languages (including Telugu, Tenglish, and Hinglish), and an automated panic-value emergency intercept.

2. Labeled Assumptions (Directly from Client Brief)

- **[ASSUMPTION-01] Patient Volume:** 2,000 patients/day over 360 operational days/year = 7,20,000 diagnostic orders annually.
- **[ASSUMPTION-02] Front-Desk Inefficiency:** 30% of patients call or visit about status (600 visits/day × 4 mins = 40 staff hours/day × 360 days = 14,400 hours × ₹250 = ₹36 Lakh/year).
- **[ASSUMPTION-03] Paper Waste:** 40% of reports printed and hand-delivered (2,88,000 reports × ₹12 = ₹34.56 Lakh). Total operational waste equals ₹70.56 Lakh annually.
- **[ASSUMPTION-04] Family Phone Sharing:** One mobile number frequently covers multi-generational families in semi-urban and rural Andhra Pradesh.

- **[ASSUMPTION-05] Statutory Retention Windows:** 72 hours for patient expiring links under DPDP data minimization; 90 days active consultation window for referring clinicians.
- **[ASSUMPTION-06] Budget & Delivery Milestone:** ₹6.5 Lakh budget across 6 weeks targeting an 80% digital adoption rate and 60% call reduction within 3 months.

3. System Architecture & Topology



4. Core Data Models

The system defines clean, structured interfaces for Orders, Users, Audit Logs, and Callback Tasks:

```
// Diagnostic Order Schema
interface DiagnosticOrder {
  orderId: string;              // e.g. "ORD-71204"
  patientName: string;          // e.g. "Lakshmi Prasanna K"
```

```
  age: number; gender: string;
  patientPhone: string;         // e.g. "+91 98765 43210"
  testName: string; parameter: string;
  result: string | number; unit: string;
  referenceRange: string; status: 'Ready' | 'Processing' | 'Critical';
  isCritical: boolean; referringDoctor: string;
  collectionCentre: string; createdAt: string;
}
```

5. Security, Cryptography & DPDP Act 2023 Compliance

- **72-Hour Signed Expiring Links:** Generated via HMAC-SHA256 signatures binding Order ID with an expiry timestamp. Tampering with parameters invalidates the cryptographic hash immediately.
- **2-Factor Mobile OTP Gate:** Entering the URL presents a 4-digit OTP verification gate sent to the registered mobile number.
- **Phone Number Masking:** Personal identifiers are automatically masked across all screens and audit logs (e.g. `+91 98765 *****`).
- **Immutable Audit Logging:** Every decryption, OTP challenge, and dispatch event is logged with timestamps and masked credentials for statutory regulatory inspection.

6. Clinical AI Mechanism & Patient Safety Safeguards

> 🚨 **The #1 Safety Rule (Critical Potassium Intercept):**
> For life-threatening results—specifically serum Potassium ≥ 6.0 mmol/L (Farzana Begum, 6.8 mmol/L):
>
> - Automated AI summaries are strictly **WITHHELD and BLOCKED**.
> - An unmissable high-contrast **Red Emergency Banner** appears directing the patient to hospital care.
> - The server automatically creates an **Urgent Doctor Callback Task** on the front-desk console requiring staff to confirm direct contact with the referring physician.

Multilingual Translation: Synthesizes plain-language summaries across 26+ Indian languages, with full support for Telugu (తెలుగు), Tenglish, Hinglish, and Hindi, backed by an in-app doubt chat assistant and a closed-loop Pathologist Audit Queue (50 summaries/week).

7. UI/UX Decisions & Clinical Accessibility

- **Medical Aurora Ambient Background:** Soft floating emerald and teal orbs with a geometric dot matrix pattern replace sterile white backgrounds.
- **Glassmorphic Elevated Cards:** High-contrast readable typography optimized for low-end mobile screens and elderly patients.
- **Search-Driven Front-Desk:** Records are dynamically revealed through intelligent searching by name, mobile, order ID, or parameter, sorted by clinical priority triage (Critical panic values first).
- **Accessible Printable Report:** Formatted to match standard pathology paper with biological reference ranges, doctor signatures, and a dedicated `@media print` PDF stylesheet.

8. What Was Cut & What We Would Do Next

What Was Cut (Per Problem Brief Out-of-Scope Constraints):

- Direct modifications to the legacy desktop LIS (solved via non-invasive CSV batch import instead).
- Online test booking and e-commerce payment gateways.
- Home collection phlebotomist GPS dispatch.

What We Would Do Next (Production Roadmap):

- Production WhatsApp Business Cloud API integration.
- ABHA (Ayushman Bharat Digital Mission) Health ID federation.
- Direct analyzer machine HL7/FHIR MLLP protocol bridges.

9. Verification & Automated Test Suite

Node.js native test runner executed on `tests/api.test.js`:

```
✓ Step 6.1 - Automated Test Suite: ArogyaLink Portal Core Services
  ✓ Security: Generate valid 72-hour signed expiring token
  ✓ Security: Reject tampered or forged HMAC tokens
  ✓ Compliance: Record and mask personal phone in audit logs
  ✓ Clinical Safety: Intercept critical panic values (Potassium 6.8)
  ✓ AI Summary: Generate multi-lingual guidance for out-of-range value (Haemoglobin 10.2)
  ✓ Auth: Authenticate Doctor with valid credentials
  ✓ Auth: Authenticate Staff with valid credentials
  ✓ Auth: Reject incorrect password
  ✓ Auth: Register new referring doctor and verify profile
  ✓ Isolation: Doctor only sees their referred patients
  ✓ LIS CSV: Parse batch CSV format correctly
11 tests passed | 0 failed | 100% pass rate
```